Analytical Chemistry Cumulative Exam
March 11, 2016
Topic: CHEM 2263

1 (30 points). You go to the lab to make up a solution of NaCl ($58.443 \pm 0.001_2$ g/mol). You weigh by difference. The weigh paper has a mass of 0.100 (±0.002) g. The mass of the solute added on top of the weigh paper is 2.634 (±0.002) g. The solute is dissolved in a volumetric flask with a volume of 100.00 (±0.08) mL.

What is the molarity of this solution and its uncertainty? Be sure to use an appropriate number of significant digits.

What part of this solution preparation process is contributing the most to the overall relative error?

2 (25 points). Describe how you would make 1 L of an acetic acid buffer if you have free access to sodium acetate and acetic acid. You may report this in moles and/or molarity rather than grams. This buffer is specified to have an ionic strength of 0.20 M and a pH of 5.0. The pK_a for acetic acid is 4.75.

3 (35 points). HPLC systems are commonly buffered. Consider the buffering capacity of 0.1 M citric acid vs. 0.1 M acetic acid buffer in the following hypothetical scenario. An HPLC column with a 150 mm (length) × 2 mm (diameter) is being buffered using one of these buffers. An undergraduate student did not listen to your instructions and injects 25 μL of a sample that contains HCl, pH 1.

a) Assume the first 1 cm of column length has 25 μL of aqueous volume. (It actually has 31.4 μL.) Assume the injected material completely mixes with the buffer in this 25 μL with no volume change (the volume is fixed at 25 μL). All acid in the injection has mixed with the buffer. What would be the pH in the first 1 cm of column length for a 0.1 M acetic acid buffer, pH 4.75?

Acetic acid: $pK_a = 4.75$

b) Assume the first 1 cm of column length has 25 μL of aqueous volume. (It actually has 31.4 μL.) Assume the injected material completely mixes in this 25 μL with no volume change (the volume is fixed at 25 μL). All acid in the injection has mixed with the buffer. What would be the pH in the first 1 cm of column length for a 0.1 M citric acid buffer, pH 4.75?

Citric Acid: $pK_{a1} = 3.1$
$pK_{a2} = 4.75$
$pK_{a3} = 6.4$

4 (10 points). In problem 3, which buffer has the better buffering capacity? Provide a chemical explanation for your choice. Bonus (10 points). Why in HPLC do we care about buffering? Does it make a difference and why?

Equation Sheet
March 11, 2016 Cumulative Exam
Analytical Chemistry

$$u_r = \sqrt{u_{x1}{}^2 + u_{x_2}{}^2 + \ldots}$$

$$\frac{u_r}{R} = \sqrt{\left(\frac{u_{x1}}{R_{x1}}\right)^2 + \left(\frac{u_{x2}}{R_{x2}}\right)^2 + \ldots}$$

$$\mu = \frac{1}{2}\sum c_i z_i{}^2$$

$$-\log \gamma_i = 0.5 z_i{}^2 \sqrt{\mu}$$